

Berribot

Engineering
Production AI Systems

Search & Ranking • LLM Evaluation • Voice Agents • AI Integrity • Cloud Infrastructure

AI Engineer • Jan–Apr 2026

42%

lower shortlisting time

Internally measured

100×1K

JD × resume benchmark

Cost / latency study

4+

services deployed

Production delivery

≈99.8%

production uptime

Production reliability

ENGINEERING NARRATIVE

Diagnosed → benchmarked → designed → built → evaluated → deployed → measured.

Owned the JD-to-candidate matching and ranking system end to end and contributed across real-time tutoring, LLM evaluation, interview integrity, internal tooling, and cloud deployment.

FLAGSHIP SYSTEM LANDSCAPE

A staged ranking architecture, not a single-model matching call.



ENGINEERING SCOPE

- Search & Match
- Candidate Intelligence
- BerriTutor
- LLM Evaluation
- Interview Integrity
- Platform / Deployment

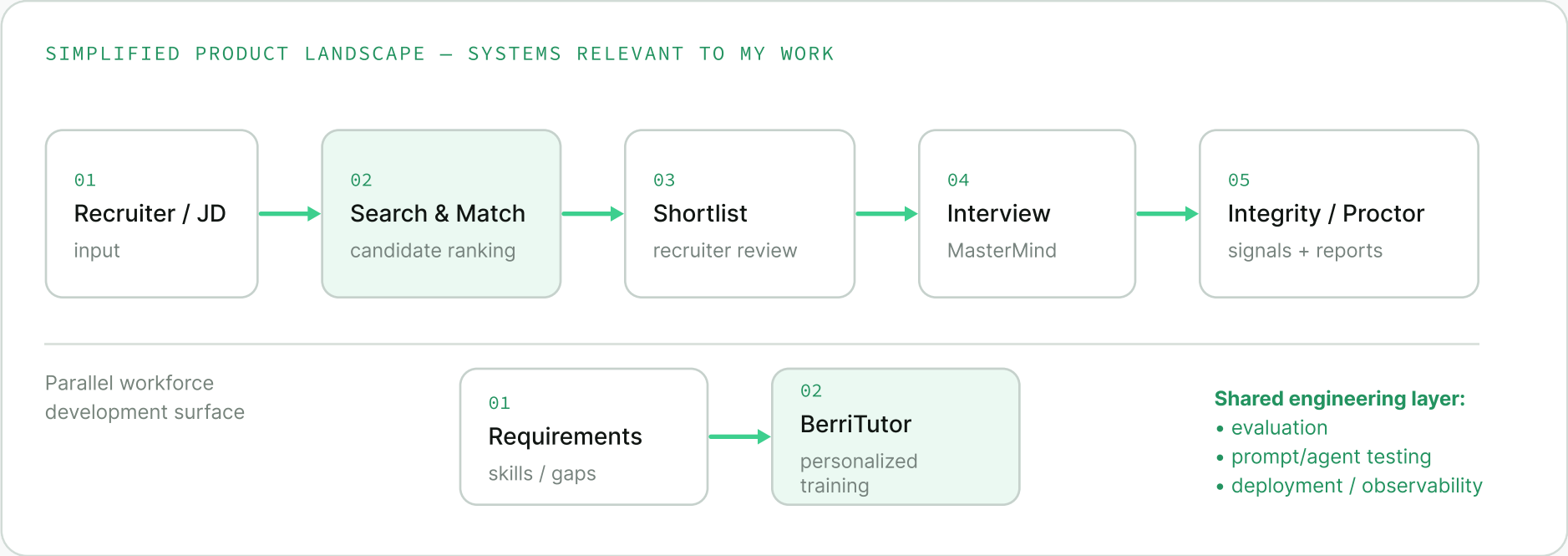
CORE TECHNOLOGIES

Python • BM25 • gemini-embedding-001 • RRF • Qwen3-Reranker • LambdaMART • W&B

LiveKit • Supermemory • MediaPipe • Docker • GitHub Actions • Artifact Registry • GKE • Cloud Run

Engineering Context & Ownership

How the systems connected — and where my ownership began and ended.



Re-architecting JD-to-Candidate Search & Ranking

Production information retrieval and learning-to-rank, from diagnosis to recruiter feedback.

BEFORE

Prior path

- Latency / cost / scalability issues
- High false-positive recommendations

INVESTIGATION

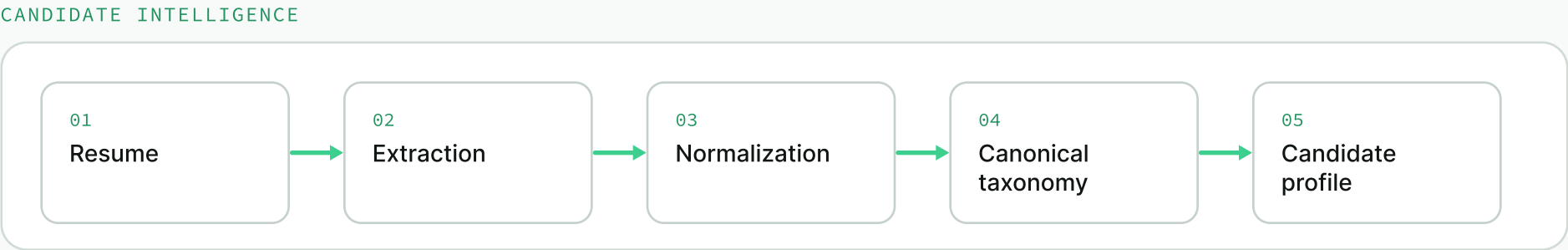
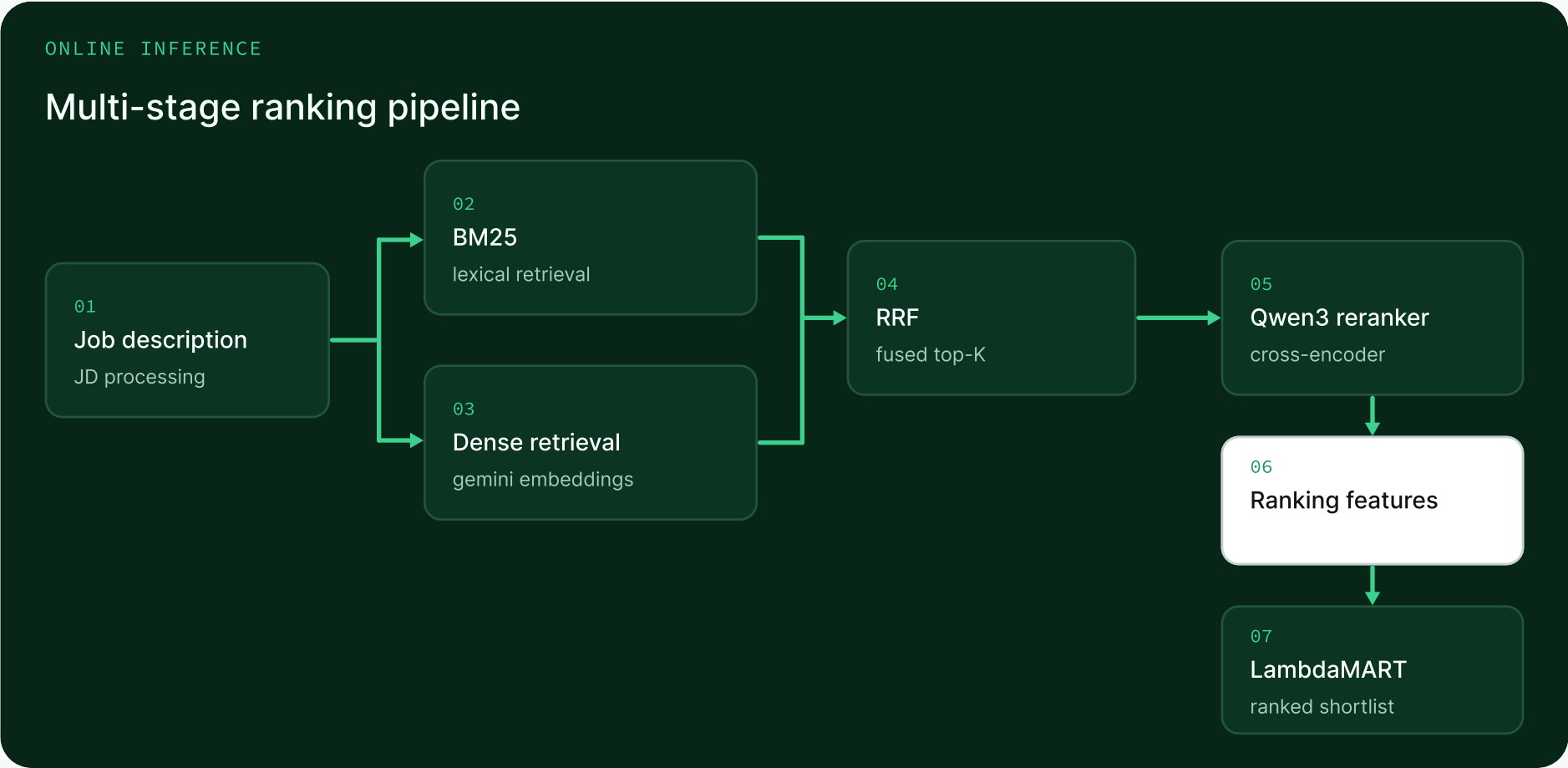
100 × 1,000

- Benchmarked 100 JD × 1,000 resume
- Exposed bottlenecks

RANKING EVALUATION

nDCG@k • MRR • Recall@k

- Recruiter judgment set
- Weights & Biases



DECISION

Hybrid retrieval

lexical + semantic coverage

DECISION

RRF

fuse unlike ranking scores

DECISION

Cross-encoder

apply expensive scoring to a reduced set

DECISION

Recruiter-labelled LTR

learn from human relevance signals

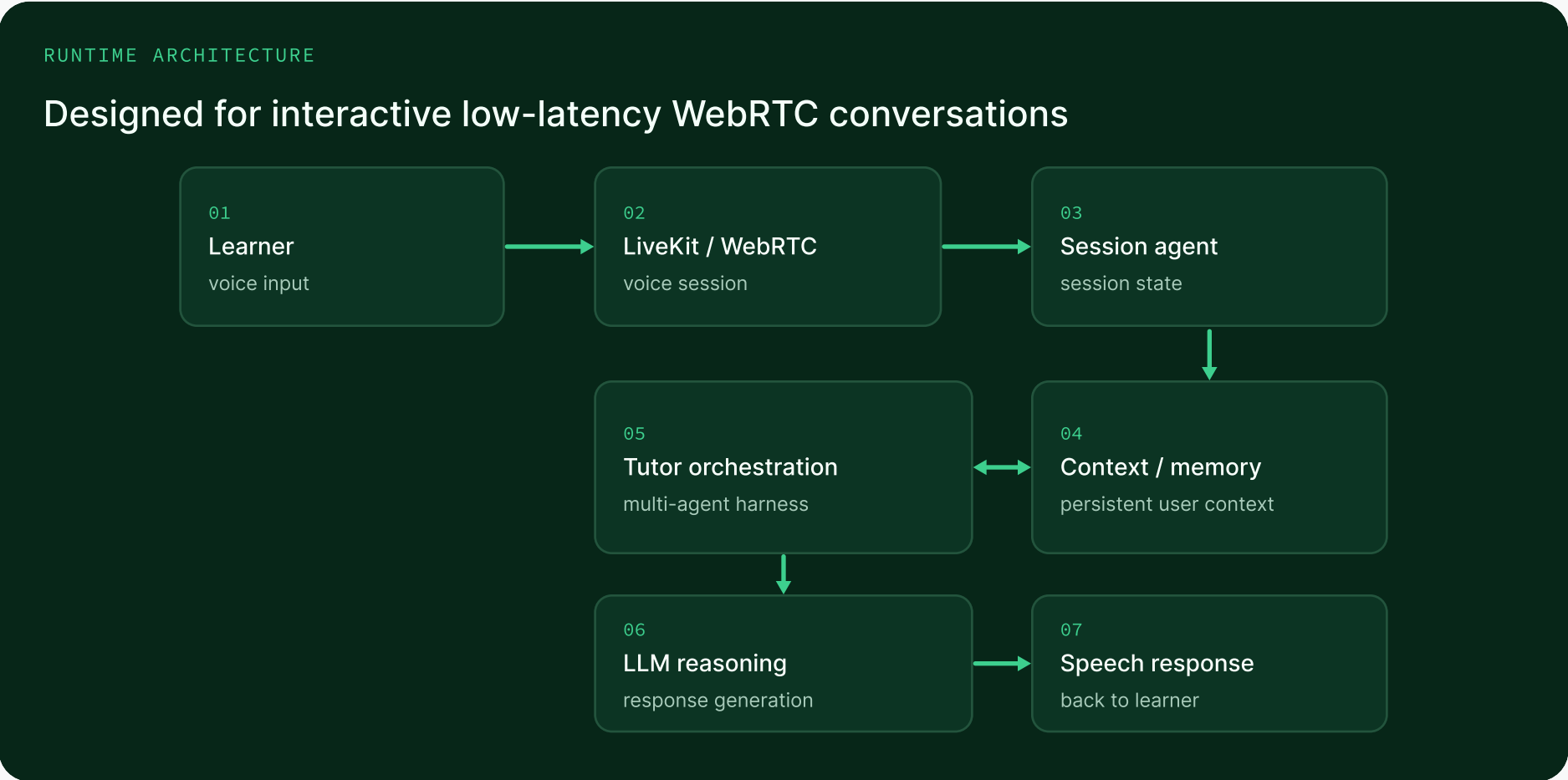
Building BerriTutor

Real-time personalized AI tutoring over voice, context, and agent orchestration.

MY ROLE

Initiated and led core development

Architecture shown at a high level to protect proprietary implementation details.



PERSONALIZATION & DELIVERY

CONTEXT

Persistent user context
User context is retrieved into the active tutoring session to personalize interaction.

TRANSPORT

Low-latency voice
LiveKit / WebRTC provides the real-time voice transport layer.

STACK

Core technology stack
Python · LiveKit · Supermemory · WebRTC · multi-agent harness

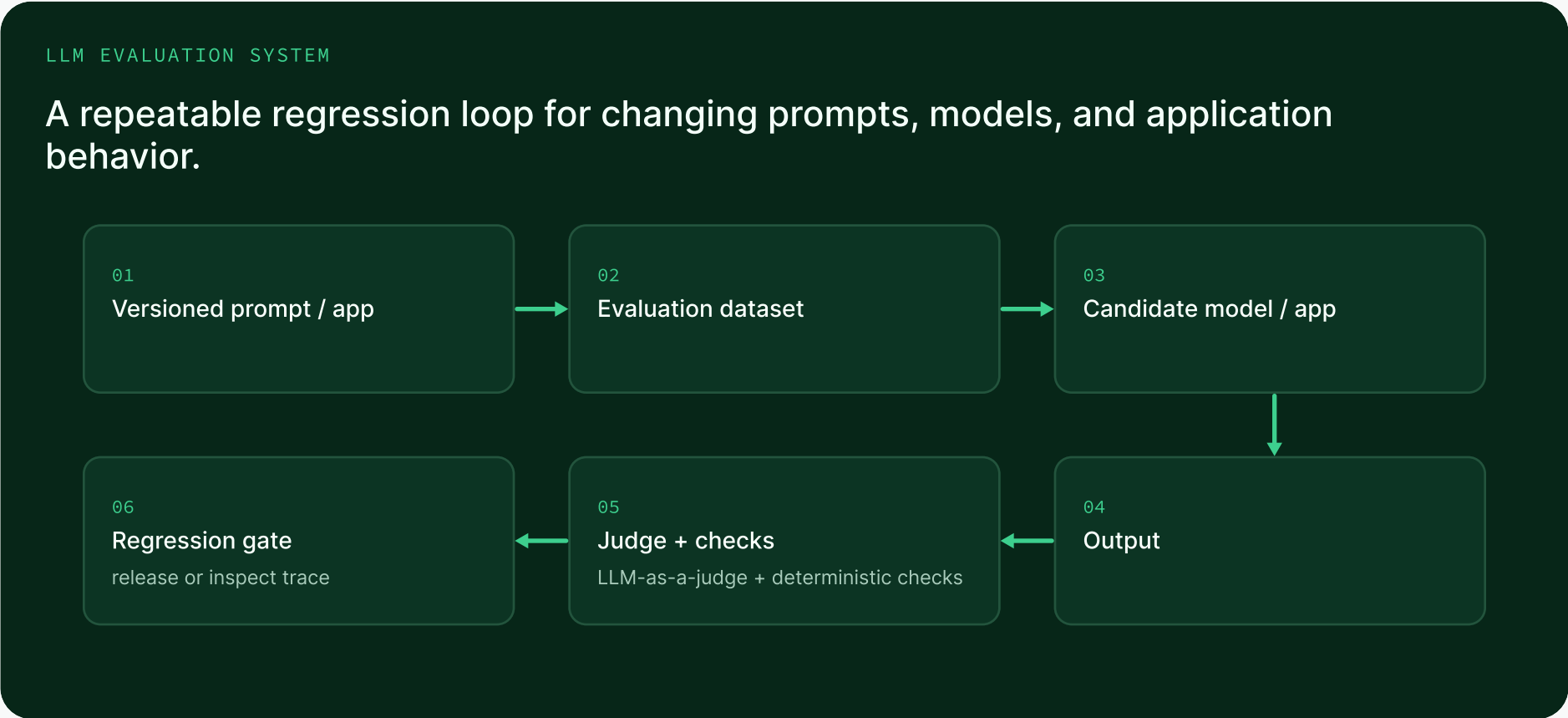
PUBLIC PRODUCT CONTEXT

Berribot publicly describes BerriTutor as personalized / adaptive training with skill-gap analysis and competency tracking.

Public product context is shown separately from proprietary implementation details.

Making AI Systems Testable, Reliable & Operable

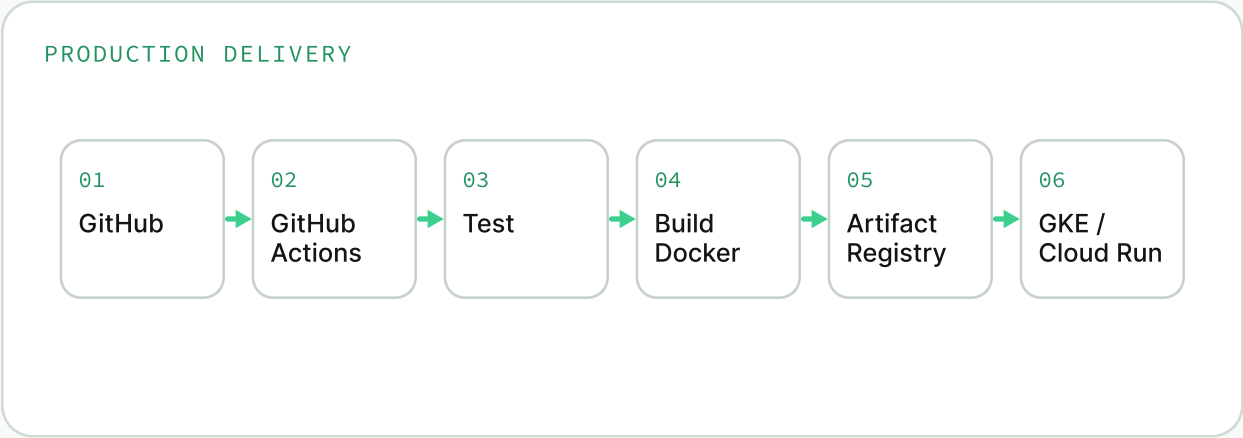
Production maturity beyond prototyping: evaluation, integrity automation, delivery, and reliability.



INTEGRITY

Interview integrity

Contributed MediaPipe-based integrity signals to help surface suspicious interview behavior.



OPERABILITY

4+

services deployed

Production scope

≈99.8%

production uptime

Production reliability

EVALUATION

Repeatable

Datasets, evaluators, comparable scores, and trace inspection.

DELIVERY

Containerized

Dockerized services delivered through GCP infrastructure.

Public artifact boundary: high-level architecture only. Proprietary code, candidate data, private endpoints, confidential prompts, and internal dashboards are excluded.

Results, Engineering Decisions & Evidence

A compact audit trail for the claims, architecture choices, and engineering lessons.

CONSOLIDATED RESULTS

SYSTEM	RESULT	FRAMING
Search / ranking	42% reduction in shortlisting time	Internal measurement
Search / ranking	100 JD × 1,000 resume benchmark	Investigation benchmark
Production	4+ services deployed	Implementation
Production	≈99.8% production uptime	Production reliability

ARCHITECTURE DECISIONS

ADR Hybrid retrieval Combine lexical and dense retrieval for complementary candidate coverage.	ADR Multi-stage ranking Retrieve broadly, then spend expensive relevance scoring on a reduced set.
ADR Recruiter-labelled LTR Use human relevance judgments rather than only fixed manual weighting.	ADR Model-independent evaluation Use shared datasets and evaluators so regression checks survive model/provider changes.

ENGINEERING LESSONS

01 Retrieval quality is architectural Lexical retrieval, semantic retrieval, reranking, LTR and evaluation work as one system.	02 Evaluation should survive model changes Application behavior matters more than coupling tests to one provider.	03 Production AI is systems engineering Latency, cost, deployment, observability and human workflows matter alongside model capability.
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EVIDENCE LANGUAGE

Public product evidence

Internal measurement

High-level architecture

Implementation